

Class 9 Gravitation – Annual Exam 2026

Full Exam-Ready Answers with Keywords

This document contains **20 highly probable questions** from the chapter **Gravitation** for the **Class 9 Annual Examination (2026)**.

Each answer is written in **full CBSE board-exam format**, followed by **keywords** to help in quick revision and marking.

A. Conceptual / Thinking Questions

1. State the Universal Law of Gravitation. Why is it called universal?

Answer:

The Universal Law of Gravitation states that every object in the universe attracts every other object with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres. This law is called *universal* because it applies to all objects in the universe, irrespective of their size or nature, ranging from atoms to planets and stars.

Keywords: attraction between all bodies, product of masses, inverse square of distance, applicable everywhere

2. Mention four phenomena explained by the Universal Law of Gravitation.

Answer:

The Universal Law of Gravitation successfully explains several natural phenomena such as: 1. The falling of objects towards the Earth. 2. The motion of the Moon around the Earth. 3. The motion of planets around the Sun. 4. The formation of tides in oceans due to the gravitational pull of the Moon and the Sun.

Keywords: falling bodies, moon's motion, planetary motion, tides

3. Differentiate between the universal gravitational constant (G) and acceleration due to gravity (g).

Answer:

The universal gravitational constant (G) is a constant that appears in the Universal Law of Gravitation and has the same value everywhere in the universe. Acceleration due to gravity (g) is the acceleration experienced by a body due to the gravitational pull of the Earth and varies from place to place.

Keywords: G constant everywhere, g varies with location, $\text{N m}^2 \text{kg}^{-2}$, m s^{-2}

4. Explain how the value of g varies from the poles to the equator.

Answer:

The value of acceleration due to gravity is greater at the poles and smaller at the equator. This is because the Earth is not a perfect sphere but slightly flattened at the poles and bulged at the equator. The radius of the Earth is smaller at the poles, which increases the value of g. Additionally, the rotation of the Earth reduces the effective value of g at the equator.

Keywords: oblate Earth, smaller radius at poles, rotation of Earth, g maximum at poles

5. State three differences between mass and weight.

Answer:

Mass is the quantity of matter contained in a body and remains constant everywhere. Weight is the force with which a body is attracted towards the Earth and varies with location. The SI unit of mass is kilogram, whereas the SI unit of weight is newton.

Keywords: quantity of matter, gravitational force, constant vs variable, kg and N

6. Define free fall. Why does the acceleration of a freely falling body not depend on its mass?

Answer:

Free fall is the motion of a body under the influence of gravitational force alone. The acceleration of a freely falling body does not depend on its mass because although the gravitational force acting on a body is proportional to its mass, the acceleration is given by force divided by mass. Hence, mass cancels out and all bodies fall with the same acceleration.

Keywords: gravity alone, force $\propto$ mass, acceleration independent of mass

7. Why is the weight of an object on the Moon only one-sixth of its weight on Earth?

Answer:

The Moon has much less mass than the Earth and therefore exerts a weaker gravitational pull. As a result, the acceleration due to gravity on the Moon is about one-sixth of that on Earth. Since weight depends on the value of g, an object weighs one-sixth on the Moon.

Keywords: smaller mass of Moon, weaker gravity, g/6

8. State Kepler's three laws of planetary motion.

Answer:

Kepler's first law states that planets move in elliptical orbits with the Sun at one focus. The second law states that the line joining a planet and the Sun sweeps out equal areas in equal intervals of time. The third law states that the square of the time period of revolution of a planet is proportional to the cube of the radius of its orbit. The gravitational force provides the necessary centripetal force for this motion.

Keywords: elliptical orbits, equal areas, $T^2 \propto r^3$, centripetal force

9. Define buoyant force and state two factors on which it depends.

Answer:

Buoyant force is the upward force exerted by a fluid on an object immersed in it. The buoyant force depends on the volume of the fluid displaced by the object and the density of the fluid.

Keywords: upward thrust, displaced fluid, volume, density

10. State Archimedes' Principle and give two applications.

Answer:

Archimedes' Principle states that when an object is immersed partially or completely in a fluid, it experiences an upward buoyant force equal to the weight of the fluid displaced by it. This principle is used in the designing of ships and submarines and in instruments such as lactometers.

Keywords: buoyant force, displaced fluid, ships, lactometer

11. What is relative density? Will an object sink if its relative density is greater than 1?

Answer:

Relative density is the ratio of the density of a substance to the density of water. If the relative density of an object is greater than 1, it will sink in water because its density is greater than that of water.

Keywords: density ratio, water, $RD > 1$ sinks

12. Why does a sheet of paper fall slower than a crumpled paper?

Answer:

A sheet of paper has a larger surface area and experiences greater air resistance compared to a crumpled paper. Due to this opposing force, the sheet of paper falls more slowly in air.

Keywords: air resistance, surface area, opposing force

B. Numerical / Application-Based Questions

13. Effect on gravitational force when distance is halved and mass doubled.

Answer:

Gravitational force is directly proportional to mass and inversely proportional to the square of distance. Doubling the mass doubles the force, and halving the distance increases the force four times. Hence, the net force becomes eight times the original force.

Keywords: $F \propto m/r^2$, $\times 2$ mass, $\times 4$ distance effect, $\times 8$ force

14. A ball is thrown vertically upward with 49 m/s. Find maximum height and total time.

Answer:

Given $u = 49 \text{ m/s}$ and $g = 9.8 \text{ m/s}^2$. At maximum height, final velocity is zero. Using equations of motion, the maximum height reached is 122.5 m. Time taken to reach maximum height is 5 s, so total time of flight is 10 s.

Keywords: equations of motion, $v = 0$, $h = 122.5 \text{ m}$, time = 10 s

15. A stone is released from a height of 19.6 m. Find its final velocity.

Answer:

The stone is dropped from rest, so initial velocity is zero. Using $v^2 = 2gh$, the final velocity just before hitting the ground is 19.6 m/s.

Keywords: free fall, $v^2 = 2gh$, 19.6 m/s

16. Find weight of a 10 kg object on Earth and Moon.

Answer:

The weight of the object on Earth is 98 N. The mass remains unchanged on the Moon, but since gravity is one-sixth, the weight on the Moon becomes approximately 16.3 N.

Keywords: $W = mg$, mass constant, Moon gravity

17. Calculate gravitational force between Earth and a 1 kg object.

Answer:

Using the Universal Law of Gravitation and given values, the gravitational force acting on the object is approximately 9.8 N.

Keywords: $F = GMm/R^2$, Earth mass, 9.8 N

18. Two stones meet problem (100 m tower).

Answer:

Using equations of motion for both stones and equating distances, the stones meet after about 2 seconds at a height of approximately 60 m from the ground.

Keywords: equations of motion, relative motion, meeting point

19. Density of substance and floating or sinking.

Answer:

The density of the substance is 2.5 g/cm^3 , which is greater than the density of water. Therefore, the substance will sink in water.

Keywords: density calculation, greater than water, sinks

20. Calculate pressure exerted by a 5 kg object.

Answer:

The force exerted by the object is 49 N. Pressure is equal to force divided by area, so the pressure exerted is 196 N/m^2 .

Keywords: pressure = force/area, thrust, 196 N/m^2

Final Board-Exam Tip

Write **definitions clearly**, start numericals with **given data**, underline **keywords**, and always mention **units**. This approach maximizes marks in CBSE evaluations.